THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 1: **Chemical Reactions and Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these is both a combination and an exothermic reaction? 1

(a) $CaCO_3 \longrightarrow CaO + CO_2$ **(b)** $2\,H_2O \longrightarrow 2\,H_2 + O_2$ **(c)** $2\,AgCl \longrightarrow 2\,Ag + Cl_2$ **(d)** $CaO + H_2O \longrightarrow Ca(OH)_2$

2. Galvanised iron does not rust because zinc: 1

(a) is less reactive than iron
(b) is lighter than iron
(c) is more reactive than iron and forms a protective layer
(d) combines with iron

3. The coefficient of CO in the balanced equation $Fe_2O_3 + CO \longrightarrow Fe + CO_2$ is: 1

(a) 3 **(b)** 1 **(c)** 4 **(d)** 2

4. In $MnO_2 + 4\,HCl \longrightarrow MnCl_2 + 2\,H_2O + Cl_2$, the substance oxidised is: 1

(a) HCl **(b)** $MnCl_2$ **(c)** MnO_2 **(d)** H_2O

5. Silver articles tarnish in air because of: 1

(a) Ag_2S formation by H_2S **(b)** Ag_2O formation by oxygen
(c) $AgCl$ formation by chlorine **(d)** melting of silver

6. In the balanced equation $C_2H_6 + O_2 \longrightarrow CO_2 + H_2O$, the coefficient of O_2 is: 1

(a) 3 **(b)** 6 **(c)** 7 **(d)** 5

7. In $CuO + H_2 \longrightarrow Cu + H_2O$, the oxidising agent is: 1

(a) CuO **(b)** H_2O **(c)** H_2 **(d)** Cu

8. Which of the following reactions is endothermic? 1

(a) Reaction of quick lime with water **(b)** Thermal decomposition of $CaCO_3$
(c) Burning of LPG **(d)** Respiration

9. The coefficient of water in the balanced equation $NH_3 + O_2 \longrightarrow NO + H_2O$ is: 1

(a) 5 **(b)** 6 **(c)** 3 **(d)** 4

10. Which metal displaces copper from copper sulphate solution? 1

(a) Fe **(b)** Au **(c)** Ag **(d)** Hg

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the blue colour of copper sulphate solution fade when an iron nail is placed in it? 2

12. Write the balanced equation for the reaction of sodium sulphate with barium chloride. State why it is called a double displacement reaction. 2

13. Why do decomposition reactions require energy? Mention two forms of energy used. 2

14. Differentiate between an oxidising agent and a reducing agent. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Balance $\mathrm{Ca(OH)_2 + HNO_3 \longrightarrow Ca(NO_3)_2 + H_2O}$ and identify the type. 3

16. Explain how rancidity can be prevented. Give three methods with reasons. 3

17. Ferrous sulphate crystals are heated in a dry test tube. Describe three changes. 3

OR

Balance $\mathrm{P_4 + O_2 \longrightarrow P_4O_{10}}$ and name the type of reaction. 3

18. An iron nail dipped in copper sulphate becomes brown, but a copper rod in ferrous sulphate shows no change. Explain. 3

OR

Silver chloride is placed in a china dish in sunlight. State the observation and give the equation and type of reaction. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe two displacement activities, one with iron nail in copper sulphate and the other with zinc in dilute acid. Give observations and equations. 5

OR

Differentiate between combination and decomposition, and between displacement and double displacement, with examples. 5

20. Explain the conditions necessary for rusting using three test tubes: dry air, boiled water with oil, and tap water. State results and prevention methods. 5

OR

A student heats green ferrous sulphate crystals. Describe the colour changes, write the equation, name the gases and state the type of reaction. 5

END OF QUESTION PAPER — Total: 40 Marks

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